

CodeWeaver comparison with SACTOR

SACTOR's Appendix Table 14 identifies an exact 50-project CRUST-Bench subset, enabling an exact-subject CodeWeaver re-analysis. All 150 CodeWeaver cells compiled; 92/150 passed every fixed project test. CodeWeaver's end-to-end project metric and static safety scan are not pooled with SACTOR's function-level, two-stage, conditionally evaluated idiomatic metric.

Denominator discipline

SACTOR evaluates 966 functions in its unidiomatic stage, then only the 32 fully successful samples in its idiomatic stage. CodeWeaver evaluates all 50 projects in every repetition against 319 fixed project tests. These denominators remain explicit.

Safety metric

CodeWeaver SafeRate is the line-weighted fraction of nonblank production Rust outside unsafe functions or blocks. SACTOR reports unsafe-free programs and average unsafe fraction with its own analyzer. The intent overlaps, but the implementations are not assumed identical.

Exact CodeWeaver measurements

Run	Build	Pass all	Fixed tests	Unsafe-free	SafeRate
CodeWeaver rep 1	50/50	31/50 (62.00%)	268/319 (84.01%)	41/50	99.83%
CodeWeaver rep 2	50/50	31/50 (62.00%)	278/319 (87.15%)	43/50	99.77%
CodeWeaver rep 3	50/50	30/50 (60.00%)	261/319 (81.82%)	40/50	99.68%

SACTOR v3 CRUST-Bench function-level comparison boundary

System	Unit	Function success	Complete samples	Unsafe-free	Note
SACTOR unidiomatic	function	81.57%	32/50 (64.00%)	0%	heavy unsafe
SACTOR idiomatic	function, conditional	42.93%	8/32 (25.00%)	100%	32 survivors only
CodeWeaver rep 1	whole project	N/A	31/50	41/50	99.83%
CodeWeaver rep 2	whole project	N/A	31/50	43/50	99.77%
CodeWeaver rep 3	whole project	N/A	30/50	40/50	99.68%

CodeWeaver execution and model-use telemetry

Scope	Cells	Elapsed h	Assistant turns	Tool calls	Premium requests	AIU	Output tokens	Input tokens
repetition 1	50	68.29	20716	32556	660	94632.82	11808699	N/A (unavailable)
repetition 2	50	64.56	19378	30588	608	87893.54	11209964	N/A (unavailable)
repetition 3	50	63.99	19108	30395	625	85477.36	10752289	N/A (unavailable)
all measured cells	150	196.85	59202	93539	1893	268003.71	33770952	N/A (unavailable)

CodeWeaver coverage measurements

Rep	Metric	Cells	Mean	Min	Max
1	coverage_before	48	64.21%	4.19%	100.00%
1	coverage_after	48	64.25%	4.19%	100.00%
2	coverage_before	49	67.94%	3.62%	100.00%
2	coverage_after	49	68.12%	3.62%	100.00%
3	coverage_before	49	64.29%	4.58%	100.00%
3	coverage_after	49	64.33%	4.58%	100.00%

CodeWeaver final-output Clippy comparison

Rep	Complete	Incomplete	Warning-free	Warnings	Errors	Alerts/project	Alerts/function
1	48	2	0	693	0	14.44	0.176
2	48	2	1	627	0	13.06	0.161
3	48	2	0	601	0	12.52	0.161

Complete source-paper surface audit

Surface	Denominator	Metrics	Artifact status
Figure 2	100 + 100 programs	unidiomatic and idiomatic SR through six attempts for five models	structured_reference
Figure 3	TransCoder-IR and CodeNet	warnings plus errors across six systems	paper_only
Table 1	TransCoder-IR and CodeNet	success, unsafe-free, average unsafe	structured_reference
Table 2	50 samples / 966 functions; conditional 32 / 580	per-sample, aggregate, full success, lint/function	structured_reference_and_codeweaver_comparison

Surface	Denominator	Metrics	Artifact status
Table 3	77 functions	success, lint/function, attempts for GPT-4o and GPT-5	structured_reference
Table 5	four datasets/codebases	size, preprocessing, tests, line/function coverage	structured_reference
Table 6	six models	version, size, temperature	documented_reference
Table 7 / Figure 7	TransCoder-IR and CodeNet failures	six and seven failure categories by model	structured_reference_categories
Table 8	successful idiomatic translations	tokens and average queries per program/model/dataset	structured_reference
Figure 8	100 programs per dataset	with/without feedback for Llama 3.3 and GPT-4o	structured_reference_summary
Table 9	50 samples / 966 functions	function/sample success and Clippy	structured_reference
Table 10	77 functions	idiomatic successes and relative drop	structured_reference
Tables 11-12	77 functions	success, Clippy, hidden unsafe dereferences, unsafe fraction	structured_reference
Figure 9	100 programs per dataset	success at temperatures 0.0, 0.5, 1.0	structured_reference_summary
Table 13	supported representation patterns	unidiomatic/idiomatic harness coverage	paper_only
Table 14	50 named samples	unidiomatic/idiomatic function success and primary failure	subset_lock_and_paper_reference